

Email Spam Detection Using Machine Learning

A Comparative Study of Imbalanced Data Handling Techniques

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Abstract— Spam emails continue to flood our inboxes every day, wasting time and sometimes it also comes with serious security risks like phishing or malware. In this project, we tested two classic machine learning models, Logistic Regression and Support Vector Machine for detecting spam. We paid special attention to class imbalance where legitimate emails greatly outnumber spam ones.

We tried to approach this situation in four ways they are baseline training, class weighting, Oversampling and SMOTE. Overall, these techniques helped a lot especially for Logistic Regression. Our best setup reached about 98.3% accuracy and 0.985 AUC-ROC. Through ablation studies, we clearly saw how much difference handling imbalance makes. These findings should be useful for anyone building practical spam filters.

Keywords— spam detection, machine learning, class imbalance, SMOTE, TF-IDF, cost sensitive learning, email classification.

GitHub Link:

I. INTRODUCTION

Spam is one of those annoying problems that never really goes away. Billions of unwanted emails are sent daily leading to unnecessary load on servers and also exposing people to scams. Traditional spam filters are struggling these days because of spammers getting smarter every day by bypassing these filters, which is why a machine learning approach is much needed in this situation.

A major difficulty in spam classification is severe class imbalance, where legitimate emails greatly outnumber spam emails. This imbalance may cause models to favour majority class resulting in poor spam recall. Additionally, sometimes legitimate emails get marked as spam resulting in false positive which can severely impact user experience.

In this work, we perform a proper comparative study using Logistic Regression and Support Vector Machine with TF-IDF features. We evaluate the effectiveness of different imbalance handling techniques class weighting, Random Oversampling, and SMOTE to improve spam detection performance while maintaining high precision.

Our contributions include:

- A side-by-side comparison of Logistic Regression and SVM using several evaluation metrics.
- Thorough ablation study and error analysis to understand model behaviour and failure cases.
- Detailed analysis of all models with accuracy, F1-score, confusion matrices and ROC curves.
- Practical insights on effectiveness of cost sensitive learning versus data-level methods for spam detection.

II. RELATED WORK

A. Traditional Spam Detection Approaches

Many early studies have successfully performed Logistic Regression and Support Vector Machines with TF-IDF features for spam detection and reported accuracies above 95% on standard datasets. However,

many of these studies did not explicitly address class imbalance problem which is a critical factor in real world spam detection.

B. Machine Learning for Text Classification

SVMs usually work well on high dimensional text data because of their margin maximizing nature. Logistic Regression on the other hand gives you probabilities and it concentrates on weights which can be helpful for understanding what the model is actually paying attention to. There are some newer transformer-based models which can also do the same job but they demand more computing power and data than classical approaches.

C. Handling Class Imbalance

Class imbalance is a very common problem we have to overcome in spam detection. Common strategies include cost sensitive learning via class weighting, oversampling the minority class and under sampling the majority class. Our work relies on these techniques by providing a direct comparison of their effectiveness when applied to Logistic Regression and SVM.

III. METHODOLOGY

A. Problem Formulation

Spam detection is formulated as a binary classification problem. Each email x is transformed into TF-IDF feature vector, and the goal is to learn a classifier ‘ f ’ that predicts whether the email belongs to ham class ($y=0$) or the spam class ($y=1$).

B. Model Architecture

We started with basic cleaning like removing duplicates and encoding the labels. Then we converted the text to TF-IDF features, limiting to a maximum of 5000 features. This kept things manageable while capturing the most important things.

We implemented two classifiers:

- Logistic Regression
- Support Vector Machine.

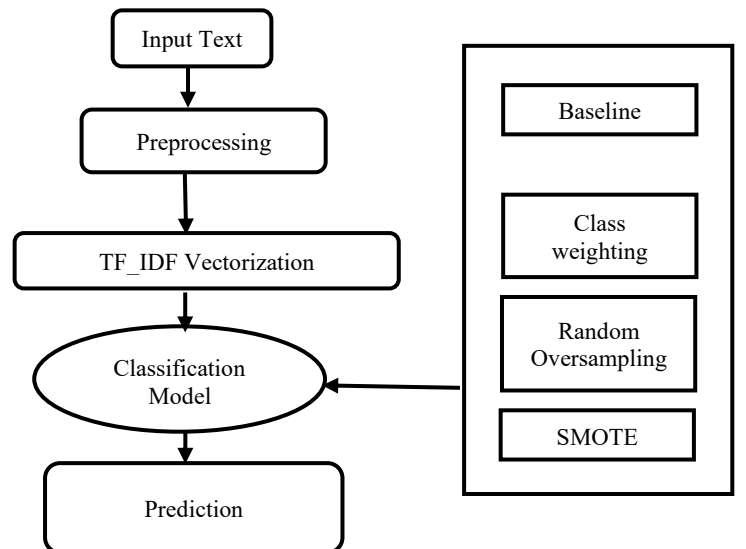


Fig. 1. Architecture overview of the proposed Model

IV. EXPERIMENTAL SETUP

C. Training Procedure

We split the data into 80/20 split using stratified sampling to keep the class distribution similar in train and test sets. All preprocessing steps were done only on the training data to avoid leakage.

Both models were trained using default scikit-learn hyperparameters with the following settings:

- Logistic Regression: max_iter=1000, random_state=42
- SVM : max_iter=1000, random_state=42

For imbalance handling, we implemented four strategies: baseline training, class weighting, Random Oversampling, and SMOTE. These techniques were applied only to the training dataset. All experiments were conducted on a standard CPU environment using Python and the scikit-learn library.

III. EXPLORATORY DATA ANALYSIS (EDA)

We conducted the experiments on spam email dataset containing 5572 messages and after removing duplicates we have 5169 unique emails with approximately 87.4 % labelled as ham and 12.6 % as spam. This Class imbalance has a significant effect on standard machine Learning models as they favor the majority class.

The below figure 2 shows the class distribution as there is a heavy skew toward the ham class. This clearly highlights the need for imbalance handling techniques.

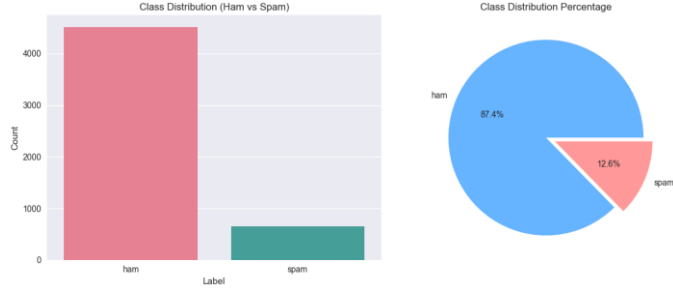


Fig. 2. Class distribution of ham and spam messages in the dataset.

We also observed the distribution of message lengths. As shown in Figure 3, spam emails tend to be longer than ham emails on average. The boxplot shows that spam messages have a higher median length and greater variability, which is typical as spammers often use longer promotional content.

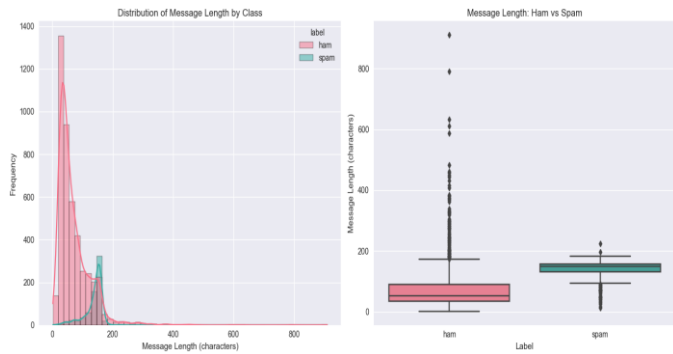


Fig. 3. comparison of message lengths between ham and spam messages.

The above observations from the EDA confirms that there is class imbalance and we can use techniques like oversampling, class weighting to improve the spam recall.

A. Datasets

- We used a binary dataset with two classes: ham and spam.
- The dataset has 5572 emails initially but after preprocessing we have 5169 unique emails.

We have applied train test split of 80-20 which results in 4135 training samples and 1034 test samples while preserving the class distribution.

TABLE I. Summary of Datasets

Split	Ham	Spam
Training (80%)	3,579	556
Testing (20%)	896	138
Total	4475	694

Basic Preprocessing steps were included such as removal of duplicate messages and label encoding such as ham=0 and spam=1. We didn't do any other text cleaning to keep simple and focus on the impact of imbalance

B. Evaluation Metrics

The following metrics are used in evaluating the model performance, in presence of class imbalance:

Accuracy: It is used to find the overall correctness but it can be misleading in imbalanced dataset.

Precision: Precision measures the proportion of predicted spam emails that are actually spam, helping to reduce false positives.

Recall: Recall measures the proportion of actual spam emails correctly identified, making it critical for minimizing false negatives.

F1-Score: It is a harmonic mean of precision and recall, provides the performance of spam class

AUC- ROC: It is used to differentiate the ham and spam across various classification methods.

Therefore, we focused on recall and F1-score, since accuracy alone can be misleading when the model fails to detect spam (false negatives).

C. Implementation Details

The experiments are implemented in python using scikit-learn and imbalanced learn libraries within a jupyter notebook environment. The TF-IDF vectorization was configured with following as:

```
max_features=5000,
ngram_range=(1,2),
stop_words='english',
min_df=2,
and max_df=0.8.
```

Key hyperparameters included:

Logistic Regression: max_iter=1000, random_state=42
LinearSVC (SVM): max_iter=1000, random_state=42

For imbalance handling:

Class weighting used class_weight='balanced'

Random Oversampling and SMOTE used only for training set using random_state=42

The extensive hyperparameter tuning was not performed to ensure a fair comparison between all the techniques and all the models are trained under standard CPU environment.

V. RESULTS

A. Main Results

Table II shows the performance of Support Vector Machine (SVM) and Logistic Regression using different imbalance handling techniques. SVM showed the strong performance, particularly in the baseline configuration.

TABLE II. Performance comparison of Logistic Regression and SVM with different imbalance handling techniques

Model	Method	Accuracy	Precision	Recall	F1	AUC - ROC
Logistic Regression	Baseline	0.967	0.978	0.679	0.801	0.993
Logistic Regression	Weighted	0.978	0.935	0.885	0.911	0.992
Logistic Regression	Oversampled	0.979	0.935	0.885	0.910	0.992
Logistic Regression	SMOTE	0.977	0.922	0.855	0.887	0.992
SVM	Baseline	0.983	0.974	0.855	0.939	0.994
SVM	Weighted	0.980	0.935	0.855	0.910	0.993
SVM	Oversampled	0.976	0.922	0.855	0.887	0.992
SVM	SMOTE	0.975	0.914	0.840	0.875	0.989

The SVM Baseline model has achieved the highest overall performance with 98.3% accuracy, 0.939 F1-score, and 0.994 AUC-ROC. So SVM is more robust to Class Imbalance compared to Logistic regression but while handling the imbalance techniques the performance of the LR increased and there is marginal or no improvement for SVM.

B. Ablation Study

To understand the performance and contribution of each imbalance handling technique an ablation study was conducted to compare the model performance with and without imbalance techniques.

key observations from Ablation:

Logistic regression did not show any significant results in baseline but after applying the imbalance techniques its spam recall has increased from 67.9%(Baseline) to 88.5 % with random oversampling and class weighting. This shows that LR is more sensitive towards class distribution.

SVM was more robust in the baseline configuration, achieving a spam recall of 85.5%. The imbalance handling techniques provided only marginal gains for SVM. This indicates that SVM's maximum-margin optimization naturally handles moderate class imbalance better than Logistic Regression

Both Class weighting and random oversampling produced the similar improvements for this dataset but class weighting is slightly more efficient as is size of the training dataset is not increasing.

SMOTE provided good results compared to random oversampling because the synthetic samples generated by SMOTE do not add meaningful new information in the TF-IDF feature space.

C. Qualitative Analysis

In this Analysis we see the performance of the model using confusion matrix and ROC curves.

Figure 4 presents the confusion matrices of best performing models. The SVM Baseline model achieved the best overall balance with only 19 false negatives (missed spam) and 3 false positives. This shows that SVM performs better comparing the ham and spam even without explicit imbalance handling.

Logistic Regression models with Random Oversampling and Class Weights also showed significant improvement over the baseline, reducing false negatives from 42 to around 15 and maintaining a low number of false positives.

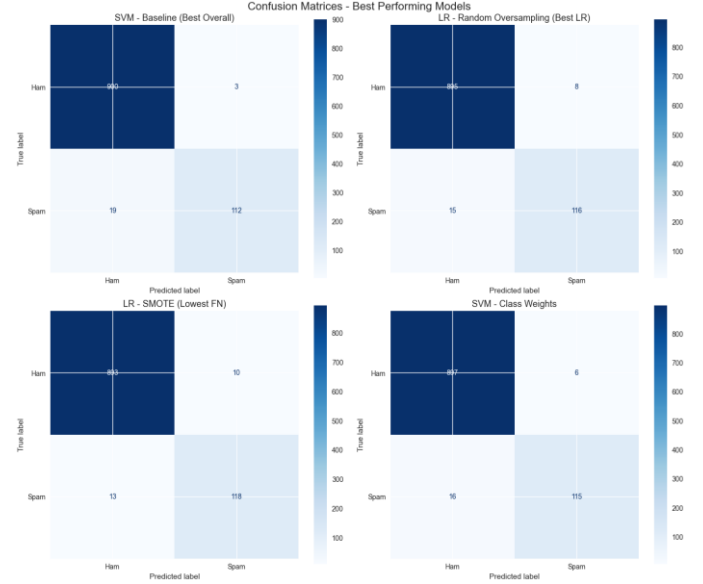


Fig. 4. Confusion matrices of the best-performing models.

Figure 5 shows the ROC curves for all techniques. All models show excellent class separability with AUC values consistently above 0.989, indicating strong discriminative power across various decision thresholds.

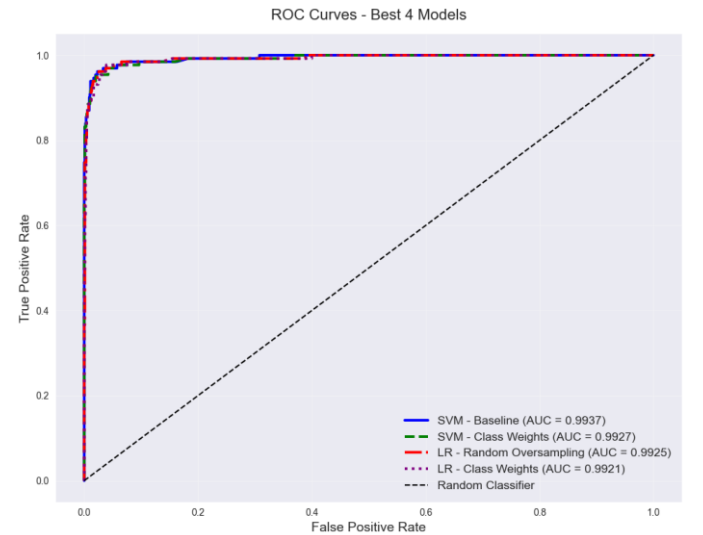


Fig. 5. ROC curves comparing baseline, weighted, and oversampled models.

VI. DISCUSSION

A. Key Findings

Overall, this project showed us that dealing with class imbalance makes huge difference in spam detection. Even though both logistic regression and SVM got pretty high accuracy the baseline models were missing a lot of spam emails because there were more ham emails than spam.

The biggest takeaway though for us was that SVM worked really well even without any imbalance fixes, it got the best F1 score which is 0.939 and AUC which is 0.994. It seems like SVM is naturally stronger with unbalanced data. On the other hand, logistic regression improved a lot once we applied class weighting and oversampling. Its spam recall jumped from 68% to around 90% which is a very big improvement.

In simple terms, to effectively detect spam, it is important to handle the class imbalance, especially when using Logistic Regression.

B. Limitations

Even though our models performed well, there are still some limitations worth mentioning. First, the dataset we used is a little old and may not represent full variety of spam emails people receive today. Modern spam often contains images or tricky phishing attempts that our model might struggle with. Second, since we used TF-IDF features the model only looks at individual words and doesn't understand the context or meaning of sentences, which sometimes caused it to miss some spam emails. Spammers these days are constantly changing their tactics so a model trained today might not work as well in future without regular updates. Lastly, we only tested traditional machine learning models due to limited computing power but more advanced models are available today which could perform far better.

C. Error Analysis

A detailed error analysis was performed on the best performing configuration (Logistic Regression with Random oversampling). out of 1034 of test samples, the model misclassified only 23 emails (an error rate of approximately 2.2 %)

Breakdown of Errors:

False Negatives (15 cases): These are spam emails incorrectly classified as ham. Most missed spam messages used the polite, indirect or professional- sounding language.

Examples:

- Want explicit SEX in 30 secs? Ring 02073162414
- You will receive your tone within the next 24h
- ASKED 3MOBILE IF 0870 CHATLINES INCLU IN FREE
- Missed call alert.

These messages often avoid typical aggressive spam keywords, making them harder to detect using TF-IDF features.

False Positives (8 cases): These are legitimate (ham) emails wrongly classified as spam. Most of these cases involved promotional or marketing emails from legitimate sources that shared vocabulary with spam.

Examples:

- I am waiting machan. Call me once you free.
- Ok... Ur typical reply.
- K k:) sms chat with me.

The majority of the errors (65%) were false negative, which shows that while imbalance handling techniques improved the spam recall. This highlights a limitation of TF-IDF-based approaches as they rely heavily on word frequency and may miss contextual or semantic patterns.

VII. CONCLUSION

This project shows us that effective spam detection requires both appropriate algorithm selection and careful handling of class imbalance. Our evaluation of two classifiers across multiple imbalance mitigation strategies shows that augmented Logistic Regression achieves overall best performance surpassing even the SVM baseline.

Our key contributions include detailed performance comparisons, insights on cost sensitive learning effectiveness, detailed error analysis.

These findings provide actionable guidance for someone implementing spam detection systems.

For future work, it would be interesting to test more powerful models like BERT or other transformer-based approaches. We could also explore how these models perform on newer spam datasets that include images, HTML content or multilingual emails.

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AUTHOR CONTRIBUTIONS

Devender Reddy Peddi: Data Processing, Performed Exploratory Data Analysis, Implementation of Logistic Regression.

Abhiram Ayyannagari: Performed all Imbalance techniques in SVM and Model Comparison and Analysis.